PINN optimizer study: actual bounded derived execution

Scope and preregistration

This is a local derived study of Rathore et al., Challenges in Training PINNs: A Loss Landscape Perspective (ICML2024), not the original study or official PaperBench. Read plan.md, inputs/TASK.md and inputs/SOURCE.md for the exact raw-coordinate architecture, data, optimizer and known earlier feasibility context. All source/assets are captured before this output; the caller's actual native receipts bind their provenance. Full requested execution=True. This document reports actual completed, failed and unexecuted cells separately and does not infer scientific success from task completion.

Mathematical and numerical methods

The raw [x,t] width100 three-hidden-layer tanh/linear-output network uses float64, prescribed Xavier initialization and the exact PDE plus concatenated-condition objective. Training consumes only residual coordinates and initial/boundary conditions, never the analytic future evaluation solution. Native Adam and one-iteration strong-Wolfe L-BFGS use the public parameters; shared11k Adam prefixes are explicit. Attempt/closure/accepted-update counts differ and remain visible. A first finite zero parameter change terminates stalled, not converged; exceptions remain real incomplete/error evidence. NNCG uses the actual shifted Nyström sketch, PCG explicit residual and Armijo sufficient descent. Delta stabilizes the projected factor, not the complete Hessian.

Actual registered results

|Configuration|State|Accepted updates|Closures|HVPs|Train loss|Relative solution error|

|---|---|---:|---:|---:|---:|---:|

|convection-adam-s345-lr1e-04|max_updates|41000|0|0|0.0025067911075800273|0.7434243060294079|

|convection-lbfgs-s345-lrnone|max_updates|41000|83632|0|0.0017896764824236625|0.6116751009888883|

|convection-hybrid-s345-lr1e-04|max_updates|41000|63039|0|2.7764376545086117e-05|0.051382167207529215|

|reaction-adam-s345-lr1e-03|max_updates|41000|0|0|5.061552633112557e-06|0.05871656407983432|

|reaction-lbfgs-s345-lrnone|max_updates|41000|117998|0|1.2727029577425702e-06|0.03201871388522498|

|reaction-hybrid-s345-lr1e-03|max_updates|41000|84327|0|9.67663801389257e-07|0.025930146954337405|

|wave-adam-s345-lr1e-03|max_updates|41000|0|0|0.02798901902881812|0.47447261723716383|

|wave-lbfgs-s345-lrnone|max_updates|41000|83124|0|0.019503473455643693|0.4669517683128822|

|wave-hybrid-s345-lr1e-03|max_updates|41000|61818|0|0.0017694103384692913|0.36803703814911914|

|convection-adam-s456-lr1e-04|max_updates|41000|0|0|0.0027146937910652916|0.7746128874670786|

|convection-lbfgs-s456-lrnone|max_updates|41000|83282|0|0.0007452011723450491|0.3002089484441467|

|convection-hybrid-s456-lr1e-04|max_updates|41000|61861|0|3.8088497106895606e-06|0.014923391260723303|

|convection-adam-s567-lr1e-04|max_updates|41000|0|0|0.003192728833540721|0.8130751782347271|

|convection-lbfgs-s567-lrnone|max_updates|41000|83497|0|0.001864624816096794|0.6325420558604921|

|convection-hybrid-s567-lr1e-04|max_updates|41000|65241|0|1.869108412711344e-06|0.01001922893110622|

|convection-adam-s345-lr1e-03|max_updates|41000|0|0|0.0003755439037324679|0.2934211862380217|

|convection-hybrid-s345-lr1e-03|max_updates|41000|61987|0|1.1309764879669062e-05|0.03435277187657833|

|convection-adam-s345-lr1e-02|max_updates|41000|0|0|0.17877097754874977|0.9999998814554625|

|convection-hybrid-s345-lr1e-02|max_updates|41000|62197|0|0.001880608207724449|0.6354986582609297|

|convection-gd-continuation-0.0001|max_updates|200|0|0|4.3079882951116675|4.326319859261342|

|convection-gd-continuation-0.001|max_updates|200|0|0|0.18333639383694594|1.0000249773827663|

|convection-nncg-continuation-0.01|algorithm_failure|0|0|421|1.1309764879669062e-05|0.03435277187657833|

|convection-nncg-continuation-0.1|max_updates|200|0|48276|8.76753054449936e-06|0.030511631290084402|

|convection-nncg-continuation-1.0|max_updates|200|0|21702|1.0490415038245618e-05|0.03374552294796272|

Same-final quality

Selection uses eligible terminal training loss only, exact-tie ID rule. The convection continuation-parent LR rule is separate. The pre-execution1% solution and0.01 residual/each-condition targets cannot be relaxed to fit these observations. Every component comes from one selected final, not a mixture. Small residual alone is insufficient: reaction initial error can amplify, and periodic/boundary defects are independent. No continuous-domain error certificate follows from the discrete public grid.

convection: {"config_id": "convection-nncg-continuation-0.1", "metrics": {"problem": "convection", "residual_points": 10000, "constraint_count": 358, "evaluation_points": 8385, "residual_loss": 1.5358777517962098e-07, "constraint_loss": 8.61394276931974e-06, "total_loss": 8.76753054449936e-06, "residual_rmse": 0.0005542342017227392, "evaluation_residual_loss": 1.8148692511197072e-07, "evaluation_residual_rmse": 0.000602473111619051, "constraint_families": {"initial_value": {"count": 257, "loss": 1.257401112898483e-07, "rmse": 0.000501478037983416 2}, "periodic_value": {"count": 101, "loss": 3.0212636661534417e-05, "rmse": 0.0077733694961109 9}}, "l2re": 0.030511631290084402, "parameter_count": 20601}, "target_components": {"relative_solution": false, "evaluation_residual": true, "initial_value": true, "periodic_value": true}}

reaction: {"config_id": "reaction-hybrid-s345-lr1e-03", "metrics": {"problem": "reaction", "residual_points": 10000, "constraint_count": 358, "evaluation_points": 8385, "residual_loss":

4.999282353178956e-07, "constraint_loss": 4.6773556607136124e-07, "total_loss": 9.67663801389257e-07, "residual_rmse": 0.000999928232742626, "evaluation_residual_loss": 6.726282417888305e-07, "evaluation_residual_rmse": 0.0011598519231253881, "constraint_families": {"initial_value": {"count": 257, "loss": 6.442171234974306e-07, "rmse": 0.0011350921755500128}, "periodic_value": {"count": 101, "loss": 1.8668632818887325e-08, "rmse": 0.00019322853215240923}}, "l2re": 0.025930146954337405, "parameter_count": 20601}, "target_components": {"relative_solution": false, "evaluation_residual": true, "initial_value": true, "periodic_value": true}}

wave: {"config_id": "wave-hybrid-s345-lr1e-03", "metrics": {"problem": "wave", "residual_points": 10000, "constraint_count": 716, "evaluation_points": 8385, "residual_loss": 6.793527109355192 5e-06, "constraint_loss": 0.001762616811359936, "total_loss": 0.0017694103384692913, "residual_rmse": 0.0036860621561105537, "evaluation_residual_loss": 1.4036928082645263e-05, "evaluation_residual_rmse": 0.005298476777838186, "constraint_families": {"initial_value": {"count": 257, "loss": 0.0004446497446988937, "rmse": 0.029821124884849455}, "initial_velocity": {"count": 257, "loss": 1.5835586588862937e-06, "rmse": 0.0017796396595301499}, "boundary_left": {"count": 101, "loss": 0.005239385808484918, "rmse": 0.1023658713486572}, "boundary_right": {"count": 101, "loss": 0.006120531795186017, "rmse": 0.11063934015698049}}, "l2re": 0.36803703814911914, "parameter_count": 20601}, "target_components": {"relative_solution": false, "evaluation_residual": true, "initial_value": false, "initial_velocity": true, "boundary_left": false, "boundary_right": false}}

Diagnostics and recovery

diagnostics contains actual selected scalar/finite-difference checks, parent full/residual/constraint Ritz estimates and same-system unpreconditioned/preconditioned solves when a complete execution provides that parent. Ritz residuals mark convergence; finite Krylov results are not the complete spectrum or a certified condition number. Recovery uses new real child PIDs and the actual500->600 Adam and100->120 selected L-BFGS saved optimizer/RNG; unavailable states or child failures are reported, never replaced by replaying a prefix. Physical shared computation, logical trajectories and postprocessing costs are distinct; phase wall times overlap inherited prefixes and cannot simply be summed.

Independent representation and physical-prior controls

The same raw width100/tanh architecture can be compiled from initial/boundary facts and PDE structure using a linear fixed-feature fit. These actual network exports, fit matrices and metrics are stored under structure-controls. They are deliberately not eligible for the old randomized-optimizer selection and cannot prove that any Adam/L-BFGS/NNCG trajectory reproduced a solution. They distinguish representation capacity from the initialization/optimization restriction. Known future evaluation solutions are used only after fitting to evaluate the exported weights, never as fitting labels. No old scientific failure is relabeled as a successful optimizer result.

convection: {"problem": "convection", "residual_points": 10000, "constraint_count": 358, "evaluation_points": 8385, "residual_loss": 2.6420627871941584e-29, "constraint_loss": 2.1701751890929998e-08, "total_loss": 2.1701751890929998e-08, "residual_rmse": 7.2691991129617e-15, "evaluation_residual_loss": 5.969160159186566e-28, "evaluation_residual_rmse": 3.4551874505405826e-14, "constraint_families": {"initial_value": {"count": 257, "loss": 1.2166484266100216e-08, "rmse": 0.00015599028345445248}, "periodic_value": {"count": 101, "loss": 4.596475960955626e-08, "rmse": 0.0003031988113748346}}, "l2re": 0.0002345174505400113, "parameter_count": 20601}

reaction: {"problem": "reaction", "residual_points": 10000, "constraint_count": 358, "evaluation_points": 8385, "residual_loss": 1.3546084210020964e-13, "constraint_loss": 3.6370135791297394e-07, "total_loss": 3.6370149337381604e-07, "residual_rmse": 5.205013777123162e-07, "evaluation_residual_loss": 1.4516319099900293e-13, "evaluation_residual_rmse": 5.388194335749276e-07, "constraint_families": {"initial_value":

{"count": 257, "loss": 5.066345763923917e-07, "rmse": 0.0010066127124096851}, "periodic_value": {"count": 101, "loss": 2.0791179087524053e-27, "rmse": 6.448438429189512e-14}}, "l2re": 0.0005250303012369046, "parameter_count": 20601}

wave: {"problem": "wave", "residual_points": 10000, "constraint_count": 716, "evaluation_points": 8385, "residual_loss": 0.0, "constraint_loss": 7.975129360782974e-07, "total_loss": 7.975129360782974e-07, "residual_rmse": 0.0, "evaluation_residual_loss": 0.0, "evaluation_residual_rmse": 0.0, "constraint_families": {"initial_value": {"count": 257, "loss": 1.8206780813463844e-06, "rmse": 0.001908233780932716}, "initial_velocity": {"count": 257, "loss": 1.2355224748823006e-29, "rmse": 4.970960621212565e-15}, "boundary_left": {"count": 101, "loss": 5.104207530511837e-07, "rmse": 0.001010367015545523}, "boundary_right": {"count": 101, "loss": 5.104207848304024e-07, "rmse": 0.001010367046998666}}, "l2re": 0.0023896013090163168, "parameter_count": 20601}

Interpretation and limits

Actual training and selected-field plots accompany this report. Cross-problem observations use seed345; only convection has paired345/456/567 and single-seed learning-rate sensitivity, so they are not19 independent statistical samples or proof of cross-problem seed robustness. All failed methods and unavailable dependencies remain in the denominator. The numerical values above, rather than an invented improvement claim, determine whether a method improved. Original Table2 convection NNCG3.63e-7 is an attributed paper value, not any result here. Current target failure is not compensated by truthful methodology. Author reference seconds and evaluator computation do not qualify effective candidate hours or statistical/professional maturity.

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Use the captured complete source/requirements, public inputs, run_study.py command and original native execution. A fresh output retrains; reading a supplied NPZ or redrawing these figures is only re-evaluation. The actual process/argv/environment appear in study-started, phase execution, attempts, recovery and caller-owned receipts. No Provider, credentials, real user process, UI or external service is involved. The present reference is not a scored fresh OpenCorvus/Codex candidate.